

1. Fermionic Spectrum

Unlike bosons, fermions satisfy the anticommutation relation:

$$\{b, b^\dagger\} = bb^\dagger + b^\dagger b = 1, \quad b^2 = b^{\dagger 2} = 0$$

Solution 1

As we have done for bosons using commutator, let us calculate

$$\{N, b\} = \{b^\dagger b, b\} = b^\dagger b^2 + b b^\dagger b = (b^\dagger b + b b^\dagger) b = \{b, b^\dagger\} b = b$$

$$\{N, b^\dagger\} = \{b^\dagger b, b^\dagger\} = b^\dagger b b^\dagger + b^\dagger b b^\dagger = b^\dagger \{b, b^\dagger\} = b^\dagger$$

Applying $\{N, b\}$ to eigenstates of N $|n\rangle$ with eigenvalue n

$$\{N, b\} |n\rangle = N b |n\rangle + b \underbrace{N |n\rangle}_{n |n\rangle} = b |n\rangle \Rightarrow N b |n\rangle = (1-n) b |n\rangle$$

$$\text{Similarly, } \{N, b^\dagger\} |n\rangle = N b^\dagger |n\rangle + b^\dagger N |n\rangle = b^\dagger |n\rangle \Rightarrow N b^\dagger |n\rangle = (1+n) b^\dagger |n\rangle$$

→ Both $b |n\rangle$ and $b^\dagger |n\rangle$ are also eigenstates with eigenvalue $1-n$

*Notice that this is only possible if allowed eigenvalues are

$$\boxed{n=0 \text{ and } n=1}$$

$$N b |1\rangle = N |0\rangle = 0 |0\rangle \quad \checkmark$$

$$N b^\dagger |0\rangle = N |1\rangle = 1 |1\rangle \quad \checkmark$$

} any other n will yield inconsistent results.

Also b is only applicable to $|1\rangle$, $b^\dagger$ is only applicable to $|0\rangle$

$$b |n\rangle = 0, \quad \forall n \in \mathbb{Z} - \{1\} \quad ; \quad b^\dagger |n\rangle = 0, \quad \forall n \in \mathbb{Z} - \{0\}$$

Solution 2

A much elegant solution lies in inspecting the square of number op.

$$N^2 = (b^\dagger b)^2 = b^\dagger b b^\dagger b = b^\dagger b (1 - b b^\dagger) = b^\dagger b - \underbrace{b^\dagger \underbrace{b b}_{b^2=0} b^\dagger}_{b^{\dagger 2}=0} = b^\dagger b = N$$

$\rightarrow N^2 = N$ means that number op. is idempotent for fermions.

$$N^2 |n\rangle = N |n\rangle \Rightarrow n^2 |n\rangle = n |n\rangle \Rightarrow n^2 - n = 0 \Rightarrow \boxed{n=0, n=1}$$

* Since number operator N counts the number of particles in a state $|n\rangle$, eigenvals 0, 1 correspond to

- $n=0$: state is empty
- $n=1$: state is occupied by 1 particle

* It is impossible to have $N \geq 2$, meaning that no two identical Fermions can occupy the same state which is exactly what "Pauli Exclusion Principle" states.

2. Hermitian Phase Operator

Susskind - Glogower phase operator: $e^{i\phi} = (aa^\dagger)^{-1/2} a$.

Solution 1

2.a) We know that we can expand analytic functions by their Taylor series around $x=0$.

$$f(x) = \sum_{k=0}^{\infty} c_k x^k \quad \text{where} \quad c_k = \frac{f^{(k)}(0)}{k!}$$

We can apply the same logic to well behaved operators

$$f(\hat{x}) = \sum_{k=0}^{\infty} c_k \hat{x}^k \Rightarrow \text{variable } x \text{ replaced by operator } \hat{x}$$

$\hookrightarrow c_k$'s are still defined i.t.o. primal function $f(x)$

$$\text{Using } f(\hat{n}) = \sum_{k=0}^{\infty} c_k \hat{n}^k \text{ and } \hat{n}|n\rangle = n|n\rangle \Rightarrow \hat{n}^k |n\rangle = n^k |n\rangle$$

$$f(\hat{n})|n\rangle = \sum_k c_k \hat{n}^k |n\rangle = \left(\sum_{k=0}^{\infty} c_k n^k \right) |n\rangle = \underline{f(n) |n\rangle} //$$

Solution 2:

Notice that Fock states form a complete orthonormal basis for Hilbert sp. and also $\hat{n}$ is hermitian. Thus, via spectral theorem, we can expand $\hat{n}$ i.t.o. its eigenbasis.

$$\hat{n} = \sum_{m=0}^{\infty} m |m\rangle \langle m|$$

* Function of an operator is then defined as $f(\hat{n}) = \sum_{m=0}^{\infty} f(m) |m\rangle \langle m|$

$$f(\hat{n})|n\rangle = \sum_m f(m) |m\rangle \langle m|n\rangle = \sum_{m=0}^{\infty} f(m) |m\rangle \delta_{mn} = \underline{f(n) |n\rangle} //$$

$$2.b) \quad e^{i\hat{\phi}} = (a a^\dagger)^{-1/2} a = (a^\dagger a + 1)^{-1/2} a = (\hat{n} + 1)^{-1/2} a$$

$$e^{i\hat{\phi}} |n\rangle = (\hat{n} + 1)^{-1/2} a |n\rangle = (\hat{n} + 1)^{-1/2} \sqrt{n} |n-1\rangle$$

Using the previous result $f(\hat{n})|n\rangle = f(n)|n\rangle$, notice that

$$(\hat{n} + 1)^{-1/2} |n-1\rangle = ((n-1) + 1)^{-1/2} |n-1\rangle = \frac{|n-1\rangle}{\sqrt{n}}$$

$$\boxed{e^{i\hat{\phi}} |n\rangle = |n-1\rangle}$$

$\Rightarrow e^{i\hat{\phi}}$ takes $|n\rangle$ states to $|n-1\rangle$ states
Then we can write it as:

$$e^{i\hat{\phi}} = \sum_{n=1}^{\infty} |n-1\rangle \langle n|$$

$$\sum_{n=1}^{\infty} e^{i\hat{\phi}} |n\rangle \langle n| = e^{i\hat{\phi}} \underbrace{\sum_{n=1}^{\infty} |n\rangle \langle n|}_{\text{completeness}} = e^{i\hat{\phi}} = \sum_{n=1}^{\infty} |n-1\rangle \langle n|$$

$\hookrightarrow$ Note that sum starts from $n=1$ since $e^{i\hat{\phi}} |0\rangle = 0$

$\hookrightarrow$ We've used the completeness of Fock states $\sum_n |n\rangle \langle n| = \mathbb{1}$

Changing the index $n \rightarrow n+1$, we find

$$\underline{e^{i\hat{\phi}} = \sum_{n=0}^{\infty} |n\rangle \langle n+1|}$$

2.c) eigenvalue eq. : $e^{i\hat{\phi}} |\psi\rangle = \lambda |\psi\rangle$

* Expanding $|\psi\rangle$ in terms of eigenbasis (number states),

$$e^{i\hat{\phi}} |\psi\rangle = \sum_{n=0}^{\infty} c_n e^{i\hat{\phi}} |n\rangle = \sum_{n=1}^{\infty} c_n |n-1\rangle + c_0 |0\rangle = \sum_{n=0}^{\infty} c_n \lambda |n\rangle$$

For $|0\rangle$: $c_1 = c_0 \lambda$

For $|1\rangle$: $c_2 = c_1 \lambda$

$\vdots$

For $|m-1\rangle$: $c_m = c_{m-1} \lambda$

Matching coeffs of all number states,

$$c_m = c_0 \lambda^m$$

$$|\psi\rangle = c_0 \sum_{n=0}^{\infty} \lambda^n |n\rangle$$

→ Since $\lambda \in \mathbb{C}$ in general, we let $c_0 \lambda = e^{i\phi}$ with constraint $-\pi < \phi < \pi$ to ensure that mapping from states $\{|\psi\rangle\}$ to ϕ is bijective.

→ Further, notice that c_0 is arbitrary, so we can always choose it to normalize $c_0 \lambda$

$$|\psi\rangle = \sum_n (e^{i\phi})^n |n\rangle = \boxed{e^{i\phi} \sum_{n=0}^{\infty} e^{in\phi} |n\rangle} \Rightarrow \text{Eigenvalues of phase op. correspond to a phase factor.}$$

* We can confirm the eigenval equation again:

$$\begin{aligned} e^{i\hat{\phi}} |e^{i\phi}\rangle &= \sum_{n=0}^{\infty} e^{in\phi} e^{i\hat{\phi}} |n\rangle = \sum_{n=0}^{\infty} e^{in\phi} \sum_{m=0}^{\infty} |m\rangle \underbrace{\langle m+1 | n \rangle}_{\delta_{m,n-1}} \\ &= \sum_{n=0}^{\infty} e^{in\phi} |n-1\rangle = \sum_{n=0}^{\infty} e^{i(n+1)\phi} |n\rangle \quad \text{since } |n-1\rangle = 0 \text{ for } n \leq 0 \\ e^{i\hat{\phi}} |e^{i\phi}\rangle &= e^{i\phi} \sum_{n=0}^{\infty} e^{in\phi} |n\rangle = e^{i\phi} |e^{i\phi}\rangle \quad \square \end{aligned}$$

2.d)

$$\frac{1}{2\pi} \int_{-\infty}^{\infty} d\phi \left| \sum_{m=0}^{\infty} e^{im\phi} |m\rangle \right| \left| \sum_{n=0}^{\infty} e^{-in\phi} \langle n| \right|$$

$$= \sum_{n,m=0}^{\infty} \underbrace{\left[\frac{1}{2\pi} \int_{-\infty}^{\infty} d\phi e^{i(m-n)\phi} \right]}_{\delta_{mn}} |m\rangle \langle n| = \underbrace{\sum_{n=0}^{\infty} |n\rangle \langle n|}_{\text{Completeness Relation}} = \mathbb{1}$$

* Notice that $|\langle e^{i\phi} | \psi \rangle|$ measures the projection of $|\psi\rangle$ onto phase eigenstate $|e^{i\phi}\rangle$. Then we can write the prob. of state $|\psi\rangle$ having phase ϕ as $P(\phi) = C |\langle e^{i\phi} | \psi \rangle|^2$ where C is a scaling constant. Normalizing

$$\int_{-\pi}^{\pi} P(\phi) d\phi = C \langle \psi | \underbrace{\left[\int_{-\pi}^{\pi} d\phi |e^{i\phi}\rangle \langle e^{i\phi}| \right]}_{2\pi} | \psi \rangle = C \cdot 2\pi \langle \psi | \psi \rangle = 1$$

$$C = \frac{1}{2\pi} //$$

$$P(\phi) = \frac{1}{2\pi} |\langle e^{i\phi} | \psi \rangle|^2 //$$

2.e) For a Fock state: $P(\phi) = \frac{1}{2\pi} |\langle e^{i\phi} | n \rangle|^2$

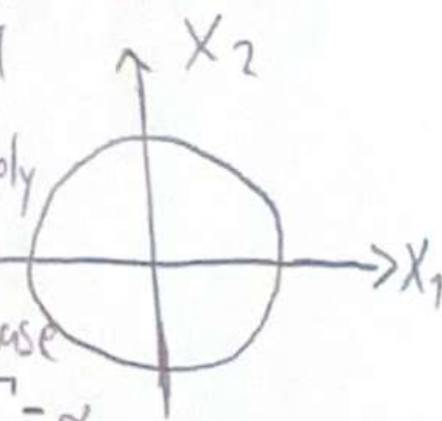
$$P(\phi) = \frac{1}{2\pi} \left| \sum_m e^{im\phi} \underbrace{\langle m | n \rangle}_{\delta_{nm}} \right|^2 = \frac{1}{2\pi} |e^{in\phi}|^2 = \frac{1}{2\pi}$$

$$\langle \phi \rangle = \int_{-\pi}^{\pi} d\phi \frac{\phi}{2\pi} = 0$$

$$\langle \phi^2 \rangle = \int_{-\pi}^{\pi} d\phi \frac{\phi^2}{2\pi} = \left[\frac{\phi^3}{6\pi} \right]_{-\pi}^{\pi} = \frac{\pi^2}{3}$$

$$\Delta\phi = \sqrt{\langle \phi^2 \rangle - \langle \phi \rangle^2} = \boxed{\frac{\pi}{\sqrt{3}}}$$

* Fock state is a infinitely thin circle on quadrature space with uniform phase prob and definite $\sqrt{n} = \alpha$



2.f) For a coherent state, similarly

$$P(\phi) = \frac{1}{2\pi} |\langle e^{i\phi} | \alpha \rangle|^2 = \frac{1}{2\pi} \left| \sum_{n=0}^{\infty} \bar{e}^{in\phi} \langle n | \cdot e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle \right|^2$$

$$= \frac{1}{2\pi} e^{-|\alpha|^2} \left| \sum_n e^{-in\phi} \frac{\alpha^n}{\sqrt{n!}} \right|^2 =$$

Let $\alpha = |\alpha| e^{i\phi}$ and taking the square-root of Gaussian approx.

$$\frac{|\alpha|^n}{\sqrt{n!}} e^{-|\alpha|^2/2} \approx (2\pi|\alpha|^2)^{-1/4} \exp\left[-\frac{(n-|\alpha|^2)^2}{4|\alpha|^2}\right] \text{ for large } |\alpha|^2$$

$$P(\phi) = \frac{1}{2\pi} \left| \sum_{n=0}^{\infty} \frac{|\alpha|^n e^{-|\alpha|^2/2}}{\sqrt{n!}} e^{-in(\phi-\phi)} \right|^2$$

$$\approx \int_0^{\infty} dn \frac{e^{-(n-|\alpha|^2)^2/4|\alpha|^2}}{(2\pi|\alpha|^2)^{1/4}} e^{-in(\phi-\phi)} \text{ for } |\alpha|^2 \gg 1$$

Let $|\alpha|^2 = \bar{n}$, $n - \bar{n} = x$, $\phi - \phi = \theta$, above expression becomes

$$\frac{e^{-i\bar{n}(\phi-\theta)}}{(2\pi\bar{n})^{1/4}} \int_0^{\infty} dx e^{-\frac{x^2}{4\bar{n}} - i\theta x}$$

$\Rightarrow$ We can extend the limit from 0 to $-\infty$ since, Gaussian dist centered at $|\alpha|^2 \gg 1$ has very little contribution from $x < 0$

$$\int_{-\infty}^{\infty} dx e^{-\left(\frac{x}{2\sqrt{\bar{n}}} + \sqrt{\bar{n}}i\theta\right)^2} e^{-\bar{n}\theta^2} = \sqrt{4\bar{n}\pi} e^{-\bar{n}\theta^2}$$

Gaussian Integral $\int_{-\infty}^{\infty} dx e^{-ax^2} = \sqrt{\frac{\pi}{a}}$

$$P(\phi) = \frac{1}{2\pi} \left(\sqrt{2} (2\pi\bar{n})^{1/4} e^{-\bar{n}\theta^2} \right)^2 |e^{-in\phi-\theta}|^2$$

$$= \frac{1}{\pi} \sqrt{2\pi\bar{n}} e^{-2\bar{n}(\phi-\phi)^2} = \left(2\pi \frac{1}{4\bar{n}}\right)^{-1/2} \exp\left[-\frac{(\phi-\phi)^2}{2 \times (1/4\bar{n})}\right]$$

$\rightarrow$ This is a gaussian with mean ϕ and variance $1/4\bar{n}$

$$\langle \phi \rangle \approx \phi, \quad \Delta\phi = \sqrt{\sigma_{\phi}^2} = \frac{1}{2\sqrt{\bar{n}}} = \frac{1}{2|\alpha|}$$

3.) Let us first confirm $\hat{a}|\alpha\rangle = \alpha|\alpha\rangle$

Using $|\alpha\rangle = e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle$ and $a|n\rangle = \sqrt{n} |n-1\rangle$

$$a|\alpha\rangle = e^{-|\alpha|^2/2} \sum_{n=1}^{\infty} \frac{\alpha^n}{\sqrt{n!}} \sqrt{n} |n-1\rangle \rightarrow \text{notice that } \frac{\sqrt{n}}{\sqrt{n!}} = \frac{1}{\sqrt{(n-1)!}}$$

$\hookrightarrow$ Notice the index change since $|n-1\rangle = 0$ for $n \leq 0$

$$a|\alpha\rangle = \alpha e^{-|\alpha|^2/2} \sum_{n=1}^{\infty} \frac{\alpha^{n-1}}{\sqrt{(n-1)!}} |n-1\rangle = \alpha e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle = \alpha|\alpha\rangle \quad \square$$

$\rightarrow$ Normalize $|\Psi\rangle = a^\dagger|\alpha\rangle$ now

$$\begin{aligned} \langle\Psi|\Psi\rangle &= \langle\alpha|aa^\dagger|\alpha\rangle = \langle\alpha|a^\dagger a + 1|\alpha\rangle = \langle\alpha|a^\dagger a|\alpha\rangle + \langle\alpha|\alpha\rangle \\ &= |\alpha\rangle^2 + 1 = |\alpha|^2 + 1 \end{aligned}$$

$$|\Psi_{\text{norm}}\rangle = \frac{a^\dagger|\alpha\rangle}{\sqrt{|\alpha|^2 + 1}}$$

Let us first calculate $\langle\hat{X}_1\rangle$ where $\hat{X}_1 = a + a^\dagger$

$$\langle\hat{X}_1\rangle = \langle\Psi_{\text{norm}}|a + a^\dagger|\Psi_{\text{norm}}\rangle = \frac{\langle\alpha|a(a + a^\dagger)a^\dagger|\alpha\rangle}{(|\alpha|^2 + 1)}$$

$$\begin{aligned} \rightarrow a(a + a^\dagger)a^\dagger &= a^2a^\dagger + aa^{\dagger 2} = a(a^\dagger a + 1) + (a^\dagger a + 1)a^\dagger \\ &= (a^\dagger a + 1)a + a + a^\dagger(a^\dagger a + 1) + a^\dagger = a^\dagger a^2 + 2a + a^{\dagger 2}a + 2a^\dagger \end{aligned}$$

$$\begin{aligned} \langle\alpha|a^\dagger a^2|\alpha\rangle &= \langle\alpha|a^\dagger a \alpha|\alpha\rangle = \alpha|\alpha|^2 \\ \langle\alpha|a^{\dagger 2}a|\alpha\rangle &= \langle\alpha|\alpha^* a^\dagger \alpha|\alpha\rangle = \alpha^*|\alpha|^2 \quad \text{and} \quad \langle\alpha|a|\alpha\rangle = \alpha \quad \text{trivially} \\ \langle\alpha|a^\dagger|\alpha\rangle &= \alpha^* \end{aligned}$$

$$\langle\hat{X}_1\rangle = \frac{(\alpha + \alpha^*)(|\alpha|^2 + 2)}{(|\alpha|^2 + 1)} //$$

In general below is a helpful relation for converting to normal ordering

$$\langle \alpha | a^n a^{\dagger m} | \alpha \rangle = \alpha^n (\alpha^*)^m + \begin{cases} n \alpha^{n-1} & , m=1 \\ m (\alpha^*)^{m-1} & , n=1 \end{cases} //$$

This can be proven easily using identity $[a, a^\dagger] = 1$ multiple times

Now let us calculate $\langle \hat{X}_2 \rangle = \langle \Psi_{\text{norm}} | (-i)(a - a^\dagger) | \Psi_{\text{norm}} \rangle$

$$\langle \hat{X}_2 \rangle = \frac{\langle \alpha | a (a - a^\dagger) a^\dagger | \alpha \rangle}{i(|\alpha|^2 + 1)} = \frac{\langle \alpha | a^2 a^\dagger - a a^{\dagger 2} | \alpha \rangle}{i(|\alpha|^2 + 1)}$$

$$= \frac{(\alpha - \alpha^*)(|\alpha|^2 + 2)}{i(|\alpha|^2 + 1)}$$

using the already calculated expressions for $a^2 a^\dagger$, $a a^{\dagger 2}$ from prev. option

Now let us calculate $\langle \hat{X}_1^2 \rangle$

$$\langle \hat{X}_1^2 \rangle = \frac{\langle \alpha | a (a + a^\dagger)^2 a^\dagger | \alpha \rangle}{|\alpha|^2 + 1}$$

$$a(a + a^\dagger)^2 a^\dagger = a^3 a^\dagger + a^2 a^{\dagger 2} + a a^\dagger a a^\dagger + a a^{\dagger 3}$$

$$\bullet \langle \alpha | a^3 a^\dagger | \alpha \rangle = \alpha^2 |\alpha|^2 + 3\alpha^2$$

$$\bullet \langle \alpha | a a^{\dagger 3} | \alpha \rangle = \alpha^{*2} |\alpha|^2 + 3\alpha^{*2}$$

$$\begin{aligned} \bullet \langle \alpha | a a^\dagger a a^\dagger | \alpha \rangle &= \langle \alpha | (a^\dagger a + 1)(a^\dagger a + 1) | \alpha \rangle = \langle \alpha | (a^\dagger a)^2 | \alpha \rangle + 2\langle \alpha | a^\dagger a | \alpha \rangle + 1 \\ &= \langle \alpha | \underbrace{a^\dagger a a^\dagger a}_{(a^\dagger a)^2 + a^\dagger a} | \alpha \rangle + 2|\alpha|^2 + 1 = |\alpha|^4 + 3|\alpha|^2 + 1 \end{aligned}$$

$$\begin{aligned} \bullet \langle \alpha | a^2 a^{\dagger 2} | \alpha \rangle &= \langle \alpha | a (a^\dagger a + 1) a^\dagger | \alpha \rangle = \underbrace{\langle \alpha | (a a^\dagger)^2 | \alpha \rangle}_{|\alpha|^4 + 2|\alpha|^2 + 1} + \underbrace{\langle \alpha | a a^\dagger | \alpha \rangle}_{|\alpha|^2 + 1} \\ &= |\alpha|^4 + 3|\alpha|^2 + 1 + |\alpha|^2 + 1 \\ &= |\alpha|^4 + 4|\alpha|^2 + 2 \end{aligned}$$

$$\langle \hat{X}_1^2 \rangle = \frac{2|\alpha|^4 + |\alpha|^2 (\alpha^2 + \alpha^{*2} + 7) + 3\alpha^2 + 3\alpha^{*2} + 3}{|\alpha|^2 + 1}$$

$$\langle \hat{X}_1 \rangle^2 = \frac{(\alpha^2 + \alpha^{*2} + 2|\alpha|^2)(|\alpha|^4 + 4|\alpha|^2 + 4)}{(|\alpha|^2 + 1)^2}$$

$$\langle \hat{X}_1^2 \rangle = \frac{2|\alpha|^6 + |\alpha|^4 (\alpha^2 + \alpha^{*2} + 9) + |\alpha|^2 (4\alpha^2 + 4\alpha^{*2} + 10) + 3\alpha^2 + 3\alpha^{*2} + 3}{(|\alpha|^2 + 1)^2}$$

$$\langle \hat{X}_1 \rangle^2 = \frac{2|\alpha|^6 + |\alpha|^4 (\alpha^2 + \alpha^{*2} + 8) + |\alpha|^2 (4\alpha^2 + 4\alpha^{*2} + 8) + 4\alpha^2 + 4\alpha^{*2}}{(|\alpha|^2 + 1)^2}$$

$$(\Delta \hat{X}_1)^2 = \frac{|\alpha|^4 + 2|\alpha|^2 + 3 - \alpha^2 - \alpha^{*2}}{(|\alpha|^2 + 1)^2} = \boxed{1 + \frac{2 - \alpha^2 - \alpha^{*2}}{(|\alpha|^2 + 1)^2}}$$

Now similarly for $\hat{X}_2$

$$\langle \hat{X}_2^2 \rangle = \frac{-\langle \alpha | a(a - a^\dagger)^2 a^\dagger | \alpha \rangle}{|\alpha|^2 + 1} = - \frac{\langle \alpha | a^3 a^\dagger + a a^\dagger^3 - a^2 a^{\dagger 2} - (a a^\dagger)^2 | \alpha \rangle}{|\alpha|^2 + 1}$$

$$= \frac{2|\alpha|^4 + |\alpha|^2 (7 - \alpha^2 - \alpha^{*2}) + 3 - 3\alpha^2 - 3\alpha^{*2}}{|\alpha|^2 + 1}$$

$$= \frac{2|\alpha|^6 + |\alpha|^4 (9 - \alpha^2 - \alpha^{*2}) + |\alpha|^2 (10 - 4\alpha^2 - 4\alpha^{*2}) + 3 - 3\alpha^2 - 3\alpha^{*2}}{(|\alpha|^2 + 1)^2}$$

$$\langle \hat{X}_2 \rangle^2 = - \frac{(\alpha^2 + \alpha^{*2} - 2|\alpha|^2)(|\alpha|^4 + 4|\alpha|^2 + 4)}{(|\alpha|^2 + 1)^2}$$

$$= \frac{2|\alpha|^6 + |\alpha|^4 (8 - \alpha^2 - \alpha^{*2}) + |\alpha|^2 (8 - 4\alpha^2 - 4\alpha^{*2}) - 4\alpha^2 - 4\alpha^{*2}}{(|\alpha|^2 + 1)^2}$$

$$(\Delta \hat{X}_2)^2 = \frac{|\alpha|^4 + 2|\alpha|^2 + 3 + \alpha^2 + \alpha^{*2}}{(1+|\alpha|^2)^2} = \boxed{1 + \frac{2 + \alpha^2 + \alpha^{*2}}{(1+|\alpha|^2)^2}}$$

• In $\alpha \rightarrow 0$ limit $\Delta \hat{X}_1 = \Delta \hat{X}_2 = \sqrt{3}$

$\Delta \hat{X}_1 \Delta \hat{X}_2 = 3 \geq 1 \Rightarrow$ Heisenberg uncertainty is satisfied

$\rightarrow \star a^\dagger |\alpha\rangle$ is definitely not a coherent state since $\Delta X_{1,2} > 1$

• In $\alpha \rightarrow \infty$ limit $\Delta \hat{X}_1 = \Delta \hat{X}_2 \approx 1$

$\rightarrow$ If the photon number is really large $a^\dagger |\alpha\rangle$ approaches to a coherent state

4) $|\psi\rangle = \frac{1}{\sqrt{2}} (|\beta\rangle + |-\beta\rangle)$ where $|\pm\beta\rangle$ are coherent

4.a) $\langle\psi|\psi\rangle = \frac{1}{2} \underbrace{\langle\beta|\beta\rangle}_{\substack{\uparrow \\ \text{coherent states} \\ \text{are normalized}}} + \frac{1}{2} \underbrace{\langle-\beta|-\beta\rangle}_{\substack{\uparrow \\ \text{coherent states} \\ \text{are normalized}}} + \frac{1}{2} \underbrace{\langle\beta|-\beta\rangle + \langle-\beta|\beta\rangle}_{\text{Re}\{\langle\beta|-\beta\rangle\}}$

$$\begin{aligned} \langle\beta|-\beta\rangle &= e^{-|\beta|^2/2} e^{-|\beta|^2/2} \sum_{m=0}^{\infty} \sum_{n=0}^{\infty} \frac{(\beta^*)^m (-\beta)^n}{\sqrt{m!} \sqrt{n!}} \langle m|n\rangle \\ &= e^{-|\beta|^2} \sum_{n=0}^{\infty} \frac{(-|\beta|^2)^n}{n!} = e^{-2|\beta|^2} \end{aligned}$$

$\hookrightarrow$ Notice we've used $\sum_{n=0}^{\infty} \frac{x^n}{n!} = e^x \Big|_{x=-|\beta|^2} = e^{-|\beta|^2}$

$\langle\psi|\psi\rangle = 1 + e^{-2|\beta|^2} \Rightarrow \text{Re}\{\cdot\}$ is dropped since $e^{-2|\beta|^2}$ is certainly real

$\lim_{|\beta|^2 \rightarrow \infty} \langle\psi|\psi\rangle = 1 \Rightarrow \langle\psi|\psi\rangle$ is normalized if $|\beta|^2 \gg 1$

4.b) Solution 1

$$\begin{aligned} |\psi\rangle &= \frac{1}{\sqrt{2}} \left[e^{-|\beta|^2/2} \sum_{n=0}^{\infty} \frac{|\beta|^n}{\sqrt{n!}} |n\rangle + e^{-|\beta|^2/2} \sum_{n=0}^{\infty} \frac{(-\beta)^n}{\sqrt{n!}} |n\rangle \right] \\ &= \sum_{n=0}^{\infty} \frac{1}{\sqrt{2}} \frac{e^{-|\beta|^2/2}}{\sqrt{n!}} [\beta^n + (-\beta)^n] |n\rangle = \sum_{n=0}^{\infty} |c_n| |n\rangle \end{aligned}$$

$$P(n) = \begin{cases} \frac{2 e^{-|\beta|^2}}{n!} |\beta|^{2n}, & n \text{ even} \\ 0, & n \text{ odd} \end{cases}$$

$\Downarrow$
Total state can be expanded in terms of number states

$P(n) = |c_n|^2$
 $\hookrightarrow$ prob. of observing $|n\rangle$

Solution 2

* Probability of observing $|n\rangle$ can be found as the inner prod.

$$P(n) = |\langle n | \psi \rangle|^2 = \frac{1}{2} \cdot \frac{e^{-|\beta|^2}}{n!} [\beta^n + (-\beta)^n]^2 \text{ similarly.}$$



* Prob. dist is still Poissonian for even n and zero for odd n .

↳ At even n 's prob. is twice the original since they take the contribution of odd n 's which were normally nonzero.

→ Odd number of photons interfere destructively and thus wavefunc. is zero, whereas even # of photons interfere constructively.

$$4.c) \quad \hat{X}(\theta) = a e^{i\theta} + a^\dagger e^{-i\theta}$$

$$\begin{aligned} \langle \hat{X}(\theta) \rangle &= \frac{1}{\sqrt{2}} (\langle \beta | + \langle -\beta |) | a e^{i\theta} + a^\dagger e^{-i\theta} | \frac{1}{\sqrt{2}} (| \beta \rangle + | -\beta \rangle) \\ &= \frac{1}{2} e^{i\theta} [\langle \beta | a | \beta \rangle + \langle -\beta | a | -\beta \rangle + \langle \beta | a | -\beta \rangle + \langle -\beta | a | \beta \rangle] \\ &\quad + \frac{1}{2} e^{-i\theta} [\langle \beta | a^\dagger | \beta \rangle + \langle -\beta | a^\dagger | -\beta \rangle + \langle \beta | a^\dagger | -\beta \rangle + \langle -\beta | a^\dagger | \beta \rangle] \end{aligned}$$

Noticing $\langle \beta | a | \beta \rangle = \langle \beta | \beta | \beta \rangle = \beta$, $\langle \beta | a^\dagger | \beta \rangle = \beta^*$

and in the limit $|\beta|^2 \xrightarrow{\text{scalar}} \infty$, $\langle \beta | -\beta \rangle = \langle -\beta | \beta \rangle \approx 0$

$$\langle \hat{X}(\theta) \rangle = \frac{1}{2} e^{i\theta} (\beta - \beta) + \frac{1}{2} e^{-i\theta} (\beta^* - \beta^*) = 0,$$

$$\langle \hat{X}^2(\theta) \rangle = \langle \Psi | e^{i2\theta} a^2 + e^{-i2\theta} a^{\dagger 2} + 2a^\dagger a + 1 | \Psi \rangle$$

$$\frac{1}{\sqrt{2}} (\langle \beta | + \langle -\beta |) | a^2 | \frac{1}{\sqrt{2}} (|\beta\rangle + |-\beta\rangle) = \frac{1}{2} [\beta^2 \underbrace{\langle \beta | \beta \rangle}_1 + (-\beta)^2 \underbrace{\langle -\beta | -\beta \rangle}_1 + \text{cross terms}]$$

$$= \beta^2$$

$$\text{Similarly } \langle \Psi | a^{\dagger 2} | \Psi \rangle = \frac{1}{2} [\beta^{*2} + (-\beta^*)^2] = |\beta^*|^2$$

$$\langle \Psi | a^\dagger a | \Psi \rangle = \frac{1}{2} [\langle \beta | \beta^* \beta | \beta \rangle + \langle -\beta | (-\beta^*) (-\beta) | -\beta \rangle] = |\beta|^2$$

$\langle \Psi | \Psi \rangle = 1$ in the $|\beta|^2 \gg 1$ limit already

$$\langle \hat{X}^2(\theta) \rangle = e^{2i\theta} \beta^2 + e^{-2i\theta} \beta^{*2} + 2|\beta|^2 + 1$$

Separate β into magnitude and phase $\Rightarrow \beta = |\beta| e^{i\phi}$

$$\langle \hat{X}^2(\theta) \rangle = 2|\beta|^2 \left[1 + \left(e^{2i(\theta+\phi)} + e^{-2i(\theta+\phi)} \right) / 2 \right] + 1$$

$$= 2|\beta|^2 [1 + \cos(2(\theta+\phi))] + 1 = 4|\beta|^2 \cos^2(\theta + \phi) + 1$$

$$\Delta X(\theta) = \sqrt{1 + 4|\beta|^2 \cos^2(\theta + \phi)}$$

* ket state is not squeezed since

$\Delta X(\theta) \geq 1$ and we need $\Delta X(\theta) < 1$ for at least some theta.

When 2 opposite coherent states are superposed, resulting states look like an ellipse $\Leftarrow$ with its major axis connecting $|\beta\rangle, |-\beta\rangle$. It has the max variance of $\sqrt{1 + 4|\beta|^2}$ in this direction and min variance 1 perpendicular to this direction

